



Research Paper

Physics-based hybrid machine learning for critical heat flux prediction with uncertainty quantification[☆]

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ARTICLE INFO

Keywords:

Critical heat flux

Dryout

Machine learning

Hybrid modeling

Uncertainty quantification

ABSTRACT

Critical heat flux (CHF) is a key quantity in nuclear system modeling due to its impact on heat transfer, safety margins, and reactor performance. This study develops and validates an uncertainty-aware hybrid modeling approach that combines machine learning with physics-based models to predict CHF in cases of dryout. The Biasi and Bowring empirical correlations were paired with three ML uncertainty quantification (UQ) techniques: deep neural network (DNN) ensembles, Bayesian neural networks (BNNs), and deep Gaussian processes (DGPs). A pure ML model without a base model was evaluated for comparison. Model performance was assessed under plentiful (7,350 points) and limited (9 points) training data scenarios using parity, uncertainty distributions, and calibration curves.

Results show that the Biasi hybrid DNN ensemble achieved the best overall performance, with a mean absolute relative error of 1.846%, and well-calibrated uncertainty estimates. The BNN-based hybrids showed slightly higher error (2.14%) but superior uncertainty calibration. DGP models underperformed, with over 6% error and poor uncertainty calibration. All hybrid models outperformed pure machine learning configurations, demonstrating resistance against data scarcity.

These findings indicate that hybrid modeling significantly improves predictive accuracy, interpretability, and resilience to data scarcity. The integration of uncertainty awareness provides actionable confidence in CHF predictions, which is vital for safety-critical decisions in nuclear applications. This hybrid approach offers a viable pathway for deploying ML models in reactor analysis tools while preserving domain knowledge and physical consistency.

1. Introduction

Critical heat flux (CHF) in a boiling system occurs when a boiling regime changes from nucleate boiling to transition boiling; this change is marked by a sudden decrease in the ability to transfer heat in a two-phase flow system. The physical presentation of CHF can generally be categorized into two mechanisms: departure from nucleate boiling (DNB) in low-quality flows, when steaming insulates the heated surface from the liquid phase, and dryout in high-quality flows, where wall film thickness approaches zero in an annular flow regime. In nuclear systems, the onset of CHF can lead to fuel cladding failure and the

potential for a fuel melt, making it a key safety-related quantity in thermal hydraulics.

Over the past several decades, researchers have developed empirical correlations and lookup tables (LUTs) to predict CHF under various thermal hydraulic conditions. Well-known models include the Biasi, Bowring, CISE-4, and W-3 correlations [1], as well as the 2006 Groeneveld LUT [2]. While these models are widely used and are computationally efficient, their accuracy degrades in regions of sparse experimental data or when extrapolated beyond calibration bounds [3]. To address these limitations, data-driven alternatives have been increasingly explored.

[☆] This manuscript has been authored by UT-Battelle LLC, under contract DE-AC05-00OR22725 with the US Department of Energy (DOE). The US government retains and the publisher, by accepting the article for publication, acknowledges that the US government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for US government purposes. DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (<http://energy.gov/downloads/doe-public-access-plan>).

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<https://doi.org/10.1016/j.applthermaleng.2025.127447>

Received 18 April 2025; Received in revised form 25 June 2025; Accepted 2 July 2025

Available online 14 July 2025

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Machine learning (ML) approaches, such as deep neural networks (DNNs), convolutional neural networks (CNNs), conditional variational autoencoders (CVAEs), support vector machines (SVMs), and random forests (RFs) have been recently applied to CHF prediction tasks [4–13]. These models are capable of learning complex input–output relationships directly from training data, and often outperform traditional models in accuracy. Recent benchmarking efforts, such as the ongoing OECD/NEA initiative for ML-based CHF prediction [14], have advanced this direction. However, most ML models are difficult to interpret and require large amounts of high-quality data to perform, which can be scarce in nuclear applications.

To address the data scarcity concern, some studies have turned to generative models to augment limited datasets [15,16]. While useful, these methods can introduce additional uncertainty and do not solve the black-box issue of interpretability. An alternative approach gaining recent interest uses a physics-based model in combination with data-driven ML model correction [17]. In this “gray-box” approach [18,19], a base model (e.g., Biasi or Bowring correlation) provides a first-estimate of the CHF value, which is then corrected by an ML model to more closely match experimental data. The bulk of physical knowledge is provided by the base model, reducing the amount of inferred knowledge required by the ML component. Structuring the model in this manner offers a larger degree of interpretability, performance benefits in comparison with stand-alone ML methods, and built-in resistance to the deleterious effects of data limitations.

Several studies have examined the use of these hybrid models. One such study compares the use of DNNs, CNNs, and RFs as the ML technique paired with the Biasi correlation, the Groeneveld LUT, or the Zuber correlation as the base model [20]. Uncertainty quantification (UQ) was performed via input perturbation to capture data uncertainty, but not uncertainty resulting from other sources. Another study [21] provides a comparison between the stand-alone LUT and the DNN, SVM, and RF hybrid counterparts. This study concluded that hybrid models exhibit greater accuracy in every case in comparison with the stand-alone LUT and non-hybrid ML variants. In addition to CHF values, the location at which CHF occurs has also been successfully predicted in rectangular channels using hybrid models [22].

In a follow-up study [23] to Zhao et al. [17], a hybrid approach implemented the Biasi and Bowring CHF empirical correlations as base models and assessed performance against a purely data-driven DNN in a series of training set sizes. The largest improvement was observed under the most data-scarce condition, where the hybrid models maintained low relative error while the pure DNN degraded significantly. Further work [24], implementing hybrid and pure ML models into the CTF subchannel code, demonstrated the usefulness of these CHF models in a production-level thermal hydraulics solver.

Despite the recent progress in data-driven modeling of CHF, two major limitations remain largely unresolved: the lack of strong uncertainty quantification and the interpretability limitations of black-box ML predictions, especially under data scarcity. Hybrid modeling approaches that combine empirical correlations with machine-learned bias corrections show promise in improving physical alignment and generalization, but have not been systematically evaluated across modern uncertainty-aware methods. This study addresses these gaps by developing and comparing three CHF prediction frameworks implementing uncertainty quantification via DNN ensembles, Bayesian neural networks (BNNs), and deep Gaussian processes (DGPs). The methodology described within this manuscript leverages residual learning in combination with low-fidelity CHF models to improve predictive performance and model uncertainties. This work supports the development of high-performing, interpretable surrogate models for use in nuclear safety analysis, where accurate CHF prediction with quantified uncertainty is critical for reliable decision-making under potentially data-limited conditions.

Table 1

Parameter ranges for the unfiltered NRC database.

Parameter	Range	
	Min	Max
D (mm)	2.00	16.00
L (m)	0.05	20.00
P (kPa)	100	20,000
G ($\text{kg m}^{-2} \text{s}^{-1}$)	8	7964
x_e (–)	–0.50	0.99
Δh_{sub} (kJ kg^{-1})	–1211	1644
T_{in} ($^{\circ}\text{C}$)	5.89	361.94
q''_{cr} (kW m^{-2})	50	16,339

Table 2

Dataset filtering criteria for adherence to both the Biasi correlation and the Bowring correlation.

Parameter	Criterion
D (m)	0.003 to 0.0375
L (m)	0.20 to 3.70
P (MPa)	0.27 to 14.0
G ($\text{kg m}^{-2} \text{s}^{-1}$)	136 to 6000
Outlet x_e	≥ 0.2

2. Background

2.1. Critical heat flux

Exceeding the CHF leads to a breakdown in heat transfer, typically presenting as dryout in boiling water reactors (BWRs). In BWRs, boiling is the primary form of heat removal from the fuel bundles. The boiling column develops from a liquid state through a series of flow regimes: bubbly, slug, and then annular, which consists of a gas core and film-coated walls. Once CHF is exceeded, the liquid film vanishes, exposing the surface and causing a significant temperature rise that can cause fuel damage. To mitigate this risk, the minimum critical power ratio is enforced to maintain adequate margin to CHF.

This study focuses exclusively on dryout conditions to ensure uniformity in CHF regime. The source dataset, derived from the public CHF database used for the 2006 Groeneveld LUT, contains both DNB and dryout experiments, which are separable by filtering on equilibrium quality (x_e). The NRC CHF database contains 24,579 points from physical experiments performed over the course of 60 years. These experiments consisted of vertical flow in uniformly heated tubes which were brought to critical conditions at the end of the heated length. The original ranges of the source database are reported in Table 1. Dryout typically occurs at higher x_e values, so a conservative threshold of $x_e > 0.2$ was applied [25].

Outside the Groeneveld LUT, two widely accepted empirical correlations to predict CHF are the Biasi [26] and Bowring [27] correlations. These correlations were selected for this study due to their widespread historical usage and are most applicable to tube geometries. Both correlations have been extensively benchmarked in open literature, have wide ranges of validity, and are well-known in thermal hydraulics [1]. Other correlations, such as the CISE-4, have more restrictive parameter constraints. Both the Biasi and Bowring correlations have unique input ranges where they are valid, so a common range for each input parameter was necessary to find a space where both correlations are valid. These ranges are listed in Table 2. All entries in the dryout dataset that fell outside these ranges were removed to ensure correlation validity. The two correlations were then applied to this filtered dataset using the heat balance method [28] and compared against known values to assess the correlations' performance in the five key metrics displayed in Table 3. These baseline results serve as reference points for assessing the benefits of the ML models developed in this work.

In Table 3, All error subscripts indicate absolute relative error, $F_{\text{error}} > 10\%$ indicates the fraction of points with absolute relative errors above 10%, and R^2 is the coefficient of determination.

Table 3

Performance metrics of the Biasi and Bowring correlations using the dryout dataset after filtering with the common constraints.

Parameter	Biasi	Bowring
μ_{error}	6.935%	6.778%
$\text{Max}_{\text{error}}$	171.8%	626.4%
$\text{Std}_{\text{error}}$	9.470%	14.22%
$F_{\text{error}} > 10\%$	19.67%	18.65%
R^2	0.9746	0.9703

2.2. UQ of data-driven machine learning

Within modeling systems, two forms of uncertainty exist: aleatoric uncertainty and epistemic uncertainty. Aleatoric uncertainty is irreducible arises from the inherent randomness present within data, from aspects such as measurement error, or from the intrinsic variability of the system being observed. Conversely, epistemic uncertainty represents the limitations of the modeling approach in attempting to capture the underlying target distribution. Epistemic uncertainty can arise from various sources, such as parameter uncertainties, differences in model architectures, and limitations in the training data. This form of uncertainty is reducible with better knowledge of the problem, larger datasets, better model architectures, and an optimized modeling technique.

In large-parameter ML approaches (e.g., DNNs), parameter uncertainty is a key epistemic consideration. Parameter uncertainty specifically arises from the randomness introduced by the stochastic nature of gradient descent and other optimization processes. One example is the initialization of weights in a DNN; initialization will affect the parameter space traversed during training, leading to different optimized weights and predictions in comparison with those of a model of a different initialization. Several techniques, such as dropout, Bayesian approaches, and ensembling, have been developed to mitigate and understand this uncertainty by averaging unique predictions or by representing weights as distributions rather than as single values. Understanding and managing these uncertainties can improve the performance and reliability of ML models. The three methods applied in this study with UQ capabilities are DNN ensembles, BNNs, and DGPs. These three approaches cover the dominant families of UQ techniques, each using a fundamentally different mechanism to capture epistemic uncertainty. They were chosen due to their conceptual diversity, demonstrated use in sensitive domains, and open-access availability. Detailed descriptions are provided in Sections 3.2, 3.3, and 3.4.

2.3. Quality of uncertainty estimates

For UQ to be actionable, the reported uncertainties must be both accurate and well-calibrated. Three aspects are considered here: calibration, magnitude, and distribution.

In the case of UQ, calibration curves are a graphical approach for assessing the *calibration* of a model's uncertainty estimates by comparing the observed cumulative distribution of standard deviation-normalized residuals with the theoretical cumulative distribution of a standard normal distribution [29]. This approach is different than the typical reliability diagram, which focuses on the calibration of predictions themselves rather than focusing on reported uncertainties. A model with well-calibrated uncertainty estimates will produce normalized residuals that closely follow a standard normal distribution, resulting in a curve that aligns with the identity line. Analyzing the calibration curve can help identify systematic overconfidence or underconfidence in the model's uncertainty estimates.

The magnitudes of uncertainty estimates in regression correspond to the model's confidence in its predictions. Uncertainty estimates are often quantified by the distribution of standard deviations from a set of predictions when given an identical input set (interpreted as samples

from an underlying distribution). If the magnitudes of these estimates are too large, then the models may lack sufficient data or be overly conservative. Overconfident predictions may also occur with small magnitudes, an indicator of unreliable uncertainty estimates.

In the consideration of the distribution of the prediction “samples”, the shape and spread of the uncertainty estimates across data points are useful for determining how a model captures variability. In regression, high-quality UQ provides varying uncertainty estimates that reflect different degrees of confidence based on the input features. This spread indicates how the model differentiates between high and low uncertainty regions, which can reflect regions of higher complexity or in extrapolation outside of the training space. Uncertainty estimates that remain constant over all input regions are neither effective nor useful.

These diagnostic tools are essential when selecting an uncertainty-aware ML model for deployment in a highly sensitive application. In practice, they support a trust-based mechanism where the uncertainty levels determine model usage. For example, if the reported certainty of a CHF prediction falls below a certain threshold, a code could revert to a physics-based model for that CHF value or issue a warning to the user. This logic is only valid when the uncertainty estimates have been rigorously validated. As such, the above diagnostics are a requirement for using uncertainty-aware ML models in safety-critical workflows.

3. Methods

The basis of the proposed hybrid approach is the prediction of residuals calculated from a base model and experimental data values. The input parameters for this problem were first selected: tube diameter (D), heated length (L), pressure (P), mass flux (G), and inlet subcooling (Δh_{sub}). The two other parameters in the original CHF dataset — inlet temperature and outlet equilibrium quality — are redundant and can be computed using the other input parameters. Therefore, these parameters may be readily excluded from this input parameter set. A base model can now be selected to provide a “first-guess” output value, which will later be corrected by the ML component. In theory, the base model can be any model that provides relevant information that links the inputs to the output value. In the case of this study, the base model corresponds to either the Biasi or Bowring correlations. Other correlations/models, such as the three-field model or even an LUT, could also be appropriate. The Groeneveld LUT was eliminated as an option for this experiment because it was constructed using the same data that were used to test and train the ML model. The use of this LUT would directly leak information during the testing phase because the holdout dataset would be polluted with values that the model could have potentially “seen” during training. This issue could lead to inflated performance scores and prevent an accurate assessment of the trained model's ability to generalize to new data.

Fig. 1 provides an overview of the hybrid model training workflow. Once the base model has been selected, the inputs of a given data entry i are fed into the model to compute $\hat{y}_i$, a traditional estimate of CHF. The difference is then taken between this value and a known experimental value y_i to compute the residual (r_i), which the ML model will be trained to predict. The data-driven model is then trained using the same set of inputs, producing a predicted output ($\hat{r}_i$). The final prediction ($\tilde{y}_i$) is given when the predicted residual is added to the base model's output to essentially provide an adjustment to the base model's first estimate, ideally closing the gap between calculated and observed values. This process is the training configuration of this approach. During actual implementation, the experimental data, and therefore the true residuals, are not known; the CHF predictions consist of only the base models' outputs combined with the ML models' predicted residuals. Although the ML component of this method predicts a different quantity than the pure ML method does (i.e., a correlation-experiment residual vs. a direct CHF value), the true output quantity of the two methods remains the same. The analysis and UQ performed in

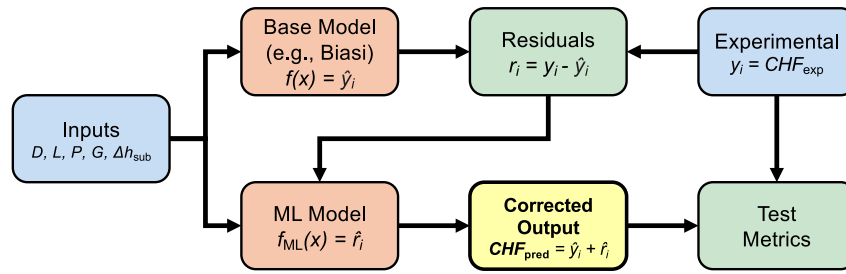


Fig. 1. Flowchart of the hybrid ML approach in the *training* configuration. Blue indicates data, pink indicates models, yellow corresponds to ancillary calculations, and green signifies the final output. In the *testing* configuration, the residuals are unknown, so it is entirely up to ML model to adjust the base model's output solely based on the input vector.

this study focused on this final CHF quantity rather than the hybrids' residuals themselves.

Three ML UQ techniques were considered in this study: DNN ensembles, BNNs, and DGPs. For each of these techniques, three choices of base models were compared: no base model, the Biasi correlation, and the Bowring correlation. These nine model configurations were trained in two data scenarios: one with plentiful training points (7350), and the other with a severely restricted training set (9 points). Using this experimental structure, an ML-base model combination may be identified as having the most favorable performance, uncertainty behavior, and resistance to performance degradation in scenarios of limited data.

3.1. Data preparation

The CHF dataset, filtered for instances of dryout within the ranges of validity for the Biasi and Bowring correlations, was first shuffled to homogenize the entries and ensure adequate coverage of the data space. The data in the input channels and the outputs were then standardized (i.e., via Z-score normalization) so that their distributions would have a mean of 0 and a standard deviation of 1 [30]. This standardization is necessary to avoid biasing the model during training, which may attribute higher importance to input channels of larger magnitude. The resulting data array was partitioned into training, validation, and testing partitions in an 80%/10%/10% split. The validation set is used for the optimization of network structure and other hyperparameters, and is separate from the testing partition, which is used to evaluate the final trained model. The use of testing data in the process of hyperparameter optimization may inadvertently transfer some knowledge of the holdout data to the model during tuning and thereby cause inflated performance metrics [31]. These data partitions and their contents were identical for every model used in this study to ensure an appropriate comparison.

A second dataset was constructed by simply reducing the training partition's size to only nine points while maintaining the size and contents of the validation and testing partitions. The same model configurations used in the plentiful data scenario would be trained on this limited data scenario. The purpose of this scenario is to investigate the hybrid approach's resistance to performance degradation due to poor training data compared to that of a pure ML model. The training set's size in this limited case represents 0.1% of the complete filtered dataset, compared with the previous scenario's training set consisting of 80%. Future development of hybrid models with limited data, for actual use, would require more comprehensive qualification.

3.2. Deep neural network ensembles

The concept of ensembling is based on the idea that a committee of unique DNN models will outperform a single model [32]. Because DNNs are purely deterministic once trained, they will produce the same value every time a prediction is made on the same input vector. When multiple unique models are well-trained with different hyperparameters and optimizations, they will each produce slightly different predictions

Table 4
DNN hyperparameter search space.

Hyperparameter	Range/Values
Hidden layers	4, 5, 6, 7, 8
Neurons	[10, 70]
Batch size	8, 16, 32, 64
Learning rate	[0.0001, 0.01]
Epochs	[200, 500]
Activation	elu, relu, softplus, sigmoid, tanh

with the same input, allowing for their treatment as samples from an underlying distribution [33]. Measures of these distributions, such as the means and standard deviations for every point in the test dataset, can then be computed. The confidence and uncertainty in the model parameters can then be quantitatively inferred from this analysis [34].

The simplest implementation of a DNN ensemble is using an initialization-based strategy, in which a set of models is trained using different weight/bias starting values [35]. Each of these models subsequently navigates a different solution space to attempt to find a global minimum in the objective function's space. In this study, a set of 20 otherwise identical models were trained with different random number generator seeds using Google's TensorFlow [36]. The network architecture consisted of seven densely connected hidden layers, using the Adam optimizer and a mean-square-error objective function. An exponential learning rate decay with a 0.96 constant was also implemented to more finely navigate the objective function's loss space in later training passes (epochs). To help further mitigate the potential for overfitting, early stopping was also used to halt training once validation loss failed to improve. The training/validation loss histories were inspected post-training to verify that both were sufficiently converged and without signs of overtraining. The network depth, as well as hyperparameters such as the number of neurons per layer and the activation functions, were optimized using the validation partition with 1000 randomized prospective configurations. RayTune [37] was used for this process, along with the asynchronous successive halving algorithm, to reduce computational expenses. The value ranges for each of the hyperparameters is reported in Table 4. Once the final architecture had been configured, the ensemble models were trained to 250 epochs and were then used to predict the outputs of the test dataset.

For each input combination in the test dataset, the 20 ensemble models produced 20 unique predictions. With these 20 samples, the means and standard deviations were collected. These mean values were used as the ensemble's "single-value" predictions in further analysis, with the standard deviations representing their uncertainties.

3.3. Bayesian neural networks

The second method, BNNs, extends the framework of DNNs but with built-in uncertainty estimation. Instead of using single-valued weights,

BNNs use probability distributions. This approach allows the model to directly capture uncertainty in the tunable parameters [38]. During training, BNNs use techniques such as variational inference to approximate the posterior distribution over the weights, providing a structured way to quantify uncertainty. Variational inference transforms the problem of exact Bayesian inference into an optimization problem, which enables the network to learn a distribution of weights that best fits the data. This allows BNNs to express uncertainty about predictions (especially in regions with limited data) and be more generalizable by considering multiple plausible models rather than a single set of weights.

The BNN was implemented using TensorFlow Probability and was constructed from four densely connected hidden layers using the Flipout estimator. These layers introduce epistemic uncertainty because they estimate a distribution over the weights and biases instead of considering them to be single values. The Flipout estimator reduces the variance in gradient estimates during training by performing a Monte Carlo approximation. The network architecture outputs two parameters defining a normal distribution: the mean and scale (standard deviation) of the target values. During training, the negative log likelihood of this distribution is minimized, which also incorporates the Kullback–Leibler divergence between prior and posterior distributions. The Kullback–Leibler weight was set as $1/N$, where N represents the number of training examples. For hyperparameter optimization, the procedure from Section 3.2 was implemented, using the same hyperparameter search space as in Table 4.

The prior and posterior distributions for the weights were modeled as normal distributions, the prior having a mean of 0 and variance of 1, and the posterior initialized with small perturbations from the prior. Variational inference was used to optimize the posterior distribution by approximating the true distribution over the weights. The model was trained to 500 epochs using the Adam optimizer with the same learning rate decay scheme as from Section 3.2. In a BNN, predictions are made from a trained model by sampling from the posterior distribution, producing unique values even when predicting on an identical input vector. For this study, 200 samples were taken from the posterior predictive distribution for every input entry in the test dataset; as with the previously discussed ensemble method, the means and standard deviations of these distributions were interpreted as the final predictions and uncertainties. The specific number of samples needed per input set was determined via a convergence study to ensure the stability of the reported values.

3.4. Deep Gaussian processes

Although DGPs also use a hierarchical network structure, they are layers of Gaussian processes (GPs). GPs are a finite collection of random variables that have a joint multivariate normal distribution. For any given input, the mean and variance are computed for a target distribution, where the mean is considered to be the prediction. GPs use kernel functions to compute the covariance between data points, making them especially useful in cases of limited data. Structuring GPs with “depth” in layers extends their ability to capture more complex representations of a dataset. Coupling this hierarchical structure with GPs allows them to effectively extract uncertainties from complex data [39]. DGPs may also be configured to capture aleatoric uncertainty via the observation noise variance within each layer, if desired. TensorFlow Probability was again used to implement the DGP models. The network architecture consisted of two layers of variational Gaussian processes using the Radial Basis Function kernel to define the covariance structure. This model was then trained to 500 epochs. The architecture and hyperparameters were tuned with the same methods of Section 3.2, but using the search space reported in Table 5.

Table 5
DGP hyperparameter search space.

Hyperparameter	Range/Values
GP Layers	1,2
Inducing points	[10, 100]
Amplitude	[0.01, 0.25]
Length scale	[0.01, 10.0]
Noise variance	[0.0001, 0.1]
Batch size	8, 16, 32, 64
Learning rate	[0.0001, 0.01]
Epochs	[200, 500]

4. Results

For each of the three ML methods used in this study, a purely data-driven case was run without a base model, and two hybrid cases were then run with the Biasi and Bowring base models. This yielded a total of nine unique model configurations, which were trained and tested in two data scenarios: plentiful training data and limited training data.

Quantifying and comparing the performance between the three modeling approaches (pure ML, Biasi hybrid, and Bowring hybrid) in each case is primarily accomplished using a set of seven metrics. The terms μ_{error} and $\text{Max}_{\text{error}}$ correspond to the mean and maximum of the absolute relative error distribution across the final predictions. The relative standard deviations (rStd) describe the shape of the distributions along each input predictions’ samples to be a measure of uncertainty for each of the final predictions. In the case of the DGP models, the standard deviations are directly extracted from the GP layers. Another error metric, relative root-mean-square error (rRMSE), is included as a more sensitive measure that penalizes larger error values to a higher degree and is scale-invariant. The definition of rRMSE has been inconsistent over several literature entries, but this study uses the definition in the OECD/NEA benchmark report, as represented in Eq. (1) [14]:

$$\text{rRMSE} (\%) = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{y}_i - y_i}{y_i} \right)^2} \times 100\% \quad (1)$$

The metric $F_{\text{error}} > 10\%$ corresponds to the percentage of absolute relative error values that are larger than 10%. Finally, as the last metric, R^2 is the coefficient of determination, which ranges from 0 to 1; an ideal model would attain a value of 1.

4.1. DNN ensembles

After all 20 ensemble models were run for each of the experimental configurations, the seven metrics were computed. These metrics are listed in Table 6, and each row’s most favorable value is highlighted in gray. For the 80% training data case, all three approaches obtain μ_{error} values below 2%, and the hybrid models present slightly smaller values in comparison with those of the pure ML approach. The maximum error values, however, are larger for the hybrid models in comparison with the pure ML model’s 36.03%. This difference is mild, and in assessing the fraction of these errors above 10%, the hybrid models are observed to have the smallest values (0.871% of the pure ML model in comparison with 0.654% and 0.764% of Biasi and Bowring, respectively). All three approaches obtain a high R^2 above 0.99, indicating strong agreement between predicted and measured values. In terms of uncertainty, all three ensembles exhibit a high degree of confidence (all below 1.55%), with the Bowring hybrid variant having the smallest relative standard (1.277%).

In the case of the 0.1% training size (using only nine points for training), the hybrid ensembles have significantly more favorable metrics in comparison with the pure ML ensemble, especially for the Biasi hybrid variant. The μ_{error} values for all three ensembles are larger than those for the 80% training size case, with the pure ML value changing from 1.947% to a far larger 35.42%. The hybrids’ μ_{error}

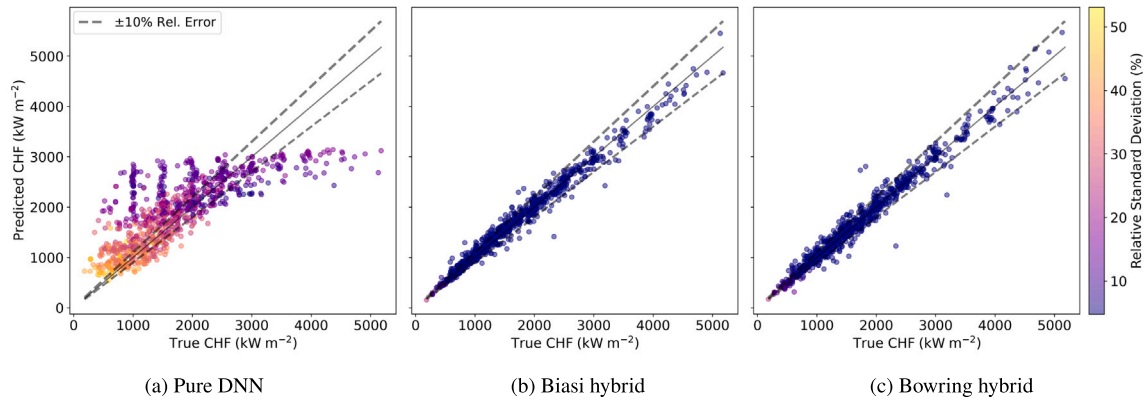


Fig. 2. Parity plots of the pure ML and hybrid ensembles with uncertainty overlay for the 0.1% training size case.

Table 6
Comparison of the ensemble performance metrics between the pure and hybrid models.

Metric	Pure ML DNN	Biasi hybrid	Bowring hybrid
7350 training points (80% case)			
μ_{error}	1.947%	1.846%	1.883%
$\text{Max}_{\text{error}}$	36.03%	37.54%	41.25%
Mean rStd	1.541%	1.302%	1.277%
Max rStd	8.967%	7.231%	11.89%
rRMSE	3.023%	2.793%	2.940%
$F_{\text{error}} > 10\%$	0.871%	0.654%	0.764%
R^2	0.9970	0.9966	0.9963
9 training points (0.1% case)			
μ_{error}	35.42%	6.082%	6.354%
$\text{Max}_{\text{error}}$	323.9%	59.10%	57.82%
Mean rStd	23.52%	1.930%	2.670%
Max rStd	53.09%	24.62%	25.43%
rRMSE	59.10%	8.804%	9.541%
$F_{\text{error}} > 10\%$	70.81%	17.10%	18.85%
R^2	0.5405	0.9782	0.9739

values increase by just under 4.5 percentage points but remain below those of the stand-alone correlations (6.935% for Biasi and 6.778% for Bowring). Compared with the stand-alone correlations, the hybrid ensembles attain more favorable values in every metric, even with only nine training points. In the case of the pure ML ensemble, the limited training data significantly impede learning and the ability to generalize to new data.

To graphically show how the three ensembles perform in the limited training data scenario, their parity plots are included in Fig. 2. These plots compare the predicted CHF value with the actual CHF value to display how “wrong” each prediction is compared with a perfect prediction along the identity line. These values are from the test dataset, which is still 10% of the original filtered dataset. The relative standard deviation on a common scale is also overlaid to show any potential trends in uncertainty.

The first behavior of note is for the pure ML ensemble, which exhibits significant deviation from the ideal line, especially in attempting to predict larger CHF values. The larger *relative* errors, however, are located in attempting to predict the smaller CHF values, as shown by the $\pm 10\%$ error bars. This larger region of relative errors also corresponds to a higher relative standard deviation, which is more favorable behavior than the converse. The most expressive comparison is between the pure ML ensemble’s parity plot and those of the hybrid ensembles. Both the Biasi and Bowring hybrids show tight adherence to the identity line; a large majority of the points are contained within the relative error bounds (only 17.10% and 18.85% lie outside for the Biasi and Bowring hybrids).

The distributions of the relative standard deviations were then plotted for the three ensemble variants in both the plentiful training data scenario and the limited scenario, as shown in Fig. 3. These plots were constructed with equal-width histograms and kernel density estimation (KDE). In the plentiful data scenario, all three ensembles perform similarly, albeit with the pure ML ensemble having a slightly higher mean and spread. Although there are larger rStd values between 7% and 12%, these are located far away from the bulk of values. In the 0.1% training size scenario, the Biasi and Bowring hybrid ensembles maintain their overall shapes with some increase in mean values, but with both mean rStd values remaining under 2.7%. The pure ML ensemble’s distribution shows a significant rise in overall rStd values; the distribution takes a bimodal shape and a maximum of 53.09%.

Evaluating the calibration of each model is also necessary. This was achieved using calibration curves and miscalibration areas. The miscalibration area is simply the area between a given curve and the identity line, which provides a quantitative aspect to this otherwise graphical analysis. Each of the three ensemble variants is plotted for both training size scenarios in Fig. 4. In the 80% case, all three of the curves adhering to the ideal line indicate that these ensembles’ predictions are well calibrated; this is further supported by relatively small miscalibration areas (0.0679, 0.0771, and 0.0824 for the pure ML, Biasi, and Bowring ensembles, respectively). In the limited data scenario, all three of the ensembles’ predictions exhibit poorer calibrations, especially in the case of Biasi. The pure ML ensemble maintains the most favorable calibration of the three, although it is entirely above the perfect calibration line. As discussed in Section 2.3, this indicates that uncertainty estimates are disproportionately large in comparison with the expected distribution, indicating underconfidence. Both of the hybrid models show the opposite effect; their estimates are smaller than expected, indicating overconfidence. Each of the miscalibration areas is larger than those of the 0.1% case: 0.113, 0.188, and 0.135 for pure ML, Biasi, and Bowring ensembles, respectively.

Both the Biasi and Bowring hybrid ensemble variants have demonstrated notable performance in accuracy and in their uncertainty estimates in comparison with those of the pure ML ensemble. Most significantly, these hybrid ensembles show a high degree of resistance to limited training datasets, achieving more favorable error metrics than even the stand-alone empirical correlations that they are based on. Although a guarantee that a hybrid approach will always perform better than or equal to that of the base model is impossible, the above results support that, at least in this case, this behavior is true.

4.2. Bayesian neural networks

The second set of cases uses BNNs to quantify the prediction uncertainties, again for the pure ML and two hybrid variants in both data scenarios. For each of the input vectors, 200 “samples” were predicted

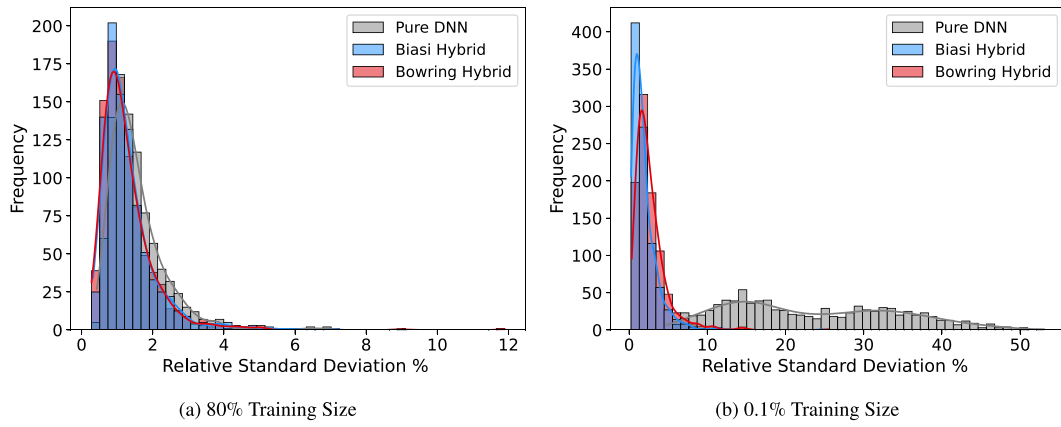


Fig. 3. Distributions of the rStds for the ensemble approach: histograms and KDE curves.

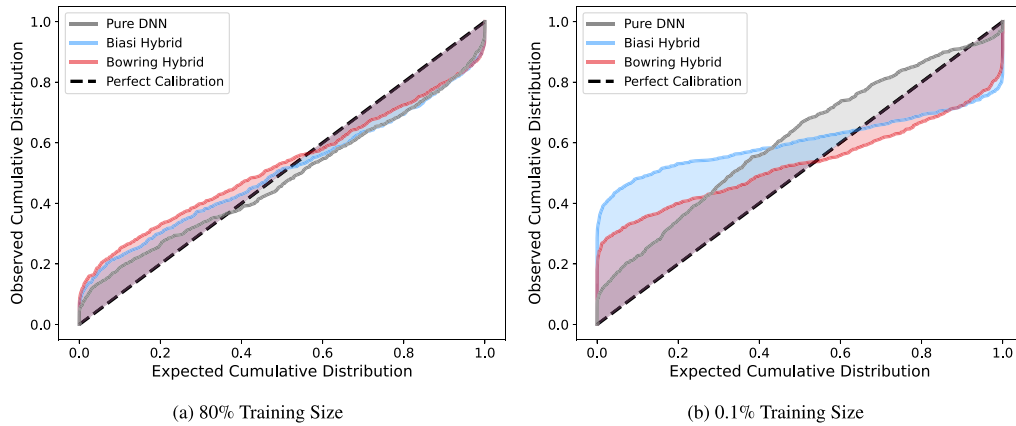


Fig. 4. Calibration curves for the DNN ensemble.

for the uncertainty estimation. The seven performance metrics were computed. These metrics are presented in Table 7. The trends noted for the DNN ensemble in Section 4.1 are present here; all three models of the 80% training size scenario behave similarly. All these values, compared with the ensemble's, are larger in error with a larger degree of uncertainty. In the limited data scenario, there is the expected jump in error in the pure ML model's predictions but with a larger degree than that of the ensemble's (2.818% up to 89.58% μ_{error}). This is also the case for the six other metrics (especially with a negative R^2), signaling a complete loss of generalization to new data. Both the Biasi and Bowring hybrid models show an expected rise in error metrics, but the Biasi hybrid exhibits values that are larger than those of the stand-alone Biasi correlation. This behavior is entirely different from the Biasi hybrid ensemble's behavior, which approached but did not exceed the error of the stand-alone correlation. The Bowring hybrid BNN model, however, remains within the performance metrics of the stand-alone correlation. Despite this decreased performance of the Biasi hybrid BNN, it reports the highest confidence in its predictions, indicating that it is not properly calibrated.

Fig. 5 presents the parity plots for these BNN models in the limited data scenario. The collapse of the pure ML model's performance is shown. The model predicts all values about a near-horizontal axis around 2250 kW m⁻². This indicates that the model fails entirely to generalize to the hidden testing dataset, effectively becoming useless in its application. The uncertainty estimates in this case are larger for smaller predicted CHF values, with a negative gradient with increasing predicted values. The two hybrid models are shown with the correct trends and relatively tight grouping around the identity line; some positive bias exists in the Biasi hybrid's predictions at lower true CHF values.

Table 7

Comparison of the BNN performance metrics between the pure and hybrid models.

Metric	Pure ML BNN	Biasi hybrid	Bowring hybrid
7350 training points (80% case)			
μ_{error}	2.818%	2.803%	2.909%
$\text{Max}_{\text{error}}$	34.28%	29.06%	38.05%
Mean rStd	3.848%	3.897%	3.833%
Max rStd	24.60%	17.27%	15.83%
rRMSE	4.109%	4.118%	4.311%
$F_{\text{error}} > 10\%$	2.503%	2.067%	2.612%
R^2	0.9926	0.9929	0.9914
9 training points (0.1% case)			
μ_{error}	89.58%	9.573%	6.474%
$\text{Max}_{\text{error}}$	1069.9%	72.91%	60.29%
Mean rStd	57.79%	12.08%	15.49%
Max rStd	72.87%	66.54%	110.5%
rRMSE	138.0%	13.13%	9.616%
$F_{\text{error}} > 10\%$	88.36%	35.80%	18.50%
R^2	-0.4934	0.9674	0.9701

To show a more focused look at the uncertainty estimates from each of these models, the relative standard deviations are provided in Fig. 6. As observed in the performance metrics, the means and shapes of the models' distributions in the plentiful training data scenario are similar. There are no significant deviations, except for the outlier just below 25%, which belongs to the pure ML model. In the limited data scenario, there is a clear difference between the distributions of the hybrid models and that of the pure ML model. The pure ML model's

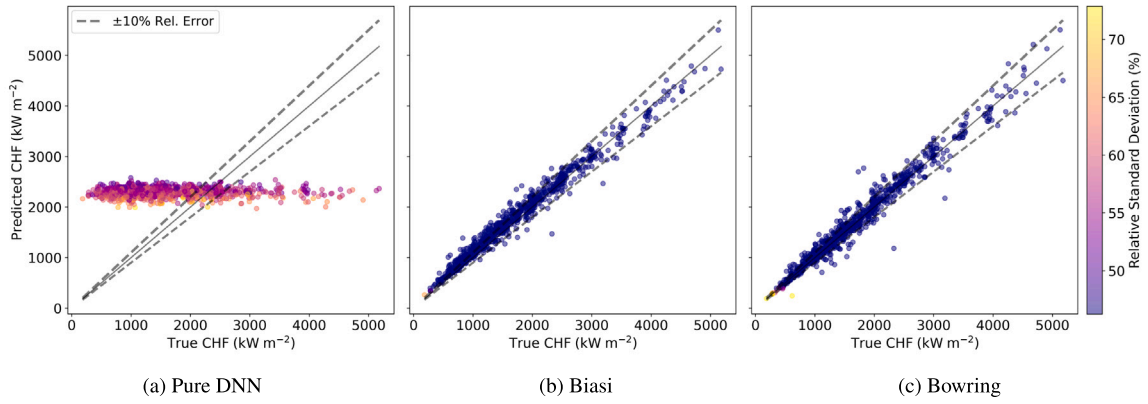


Fig. 5. Parity plots of the pure ML and hybrid BNN with uncertainty overlay for the 0.1% training size case.

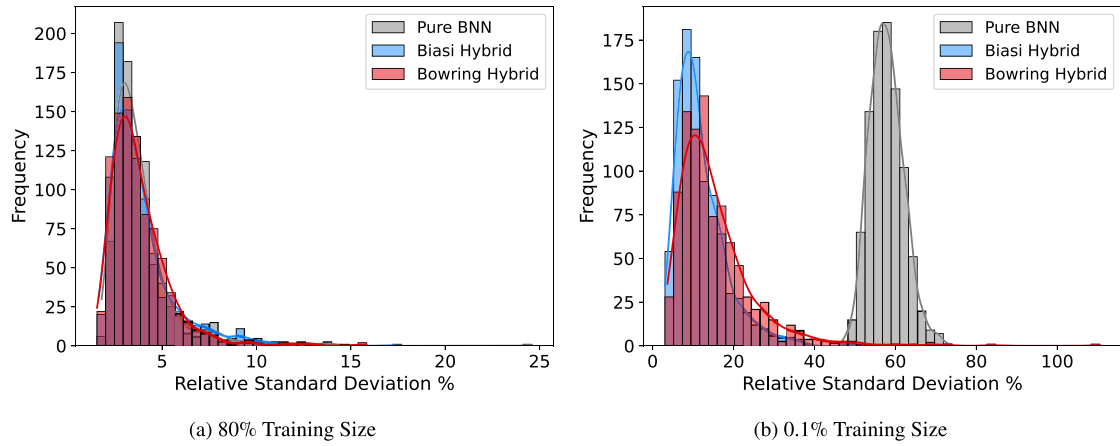


Fig. 6. Distributions of the rStd for the BNN approach: histograms and KDE curves.

rStd values are near symmetric at about the 57.79% mean value and are a distinct peak in comparison with the hybrid model distributions in the left-hand corner. This behavior is favorable to some degree because the very large error associated with this model is paired with non-confidence in its predictions.

Fig. 7 shows the calibration curves for the BNN approach. Compared with the DNN ensemble predictions in the plentiful training scenario (discussed in Section 4.1), the BNN predictions in the plentiful training scenario are extremely well calibrated; the hybrid models visibly align with the identity line. The pure ML model also indicates adequate calibration, having a miscalibration area of 0.0506 (compared with the ensemble counterpart's 0.0679). Both hybrid models' curves also feature smaller miscalibration areas: 0.0233 and 0.0115 for the Biasi and Bowring variants, respectively. This behavior is starkly contrasted by that of the limited data scenario, with miscalibration areas of 0.1691, 0.1669, and 0.1023 for the pure ML, the Biasi hybrid, and the Bowring hybrid, respectively. All three of the model variants systematically overestimate uncertainties, leading to large prediction intervals (under-confidence). The Bowring hybrid has the most favorable calibration of the three, albeit with significant underconfidence.

4.3. Deep Gaussian processes

Each of the DGP models was trained and tested using the same test dataset used for the other models. The standard deviations were extracted directly from the model instead of using a sampling approach such as the approaches used for the DNN ensembles and the BNN models. The same performance metrics were then computed. These metrics are summarized in Table 8. In comparison with the performance of the

previous two methods, nearly every error metric is significantly larger. Between the DGP models themselves, the Biasi hybrid model outperforms both the pure ML and Bowring hybrid models by a significant margin. Although it does not match the performance of the Biasi hybrid, the Bowring hybrid model also performs with more favorable metrics in comparison with the pure ML model. The exception in this case is in the Bowring hybrid's larger relative standard deviation values, which indicate that more uncertainty is associated with those predictions.

In the limited training dataset case (using only nine points for training), all three of the models' predictions were observed with expected deterioration in all metrics. The two hybrid models have notably smaller error metrics compared with those of the pure ML model, except for the maximum relative standard deviation. Of the three models, the Bowring hybrid produced the most accurate predictions with the smallest uncertainties on average. Notably, as with the pure BNN models, the pure DGP predicts nearly the same value for every input combination, indicating a complete inability to generalize to new data.

The parity plots with uncertainty overlays were once again generated for the limited training dataset case. These plots are nearly identical in behavior to those of the BNN, showing a complete deterioration of the pure ML model's predictions with a similar relative standard deviation spread. Because of this redundancy, these parity plots are omitted from this section.

The relative standard deviations for each of the points were extracted and plotted to assess the behavior of this distribution, as shown in Fig. 8. Because of the large maximum rStd values in the hybrid models (681.9% for Biasi and 218.6% for Bowring in the limited training data case), all outliers are removed from Fig. 8 for visual purposes. In the plentiful training dataset case, the pure ML model has

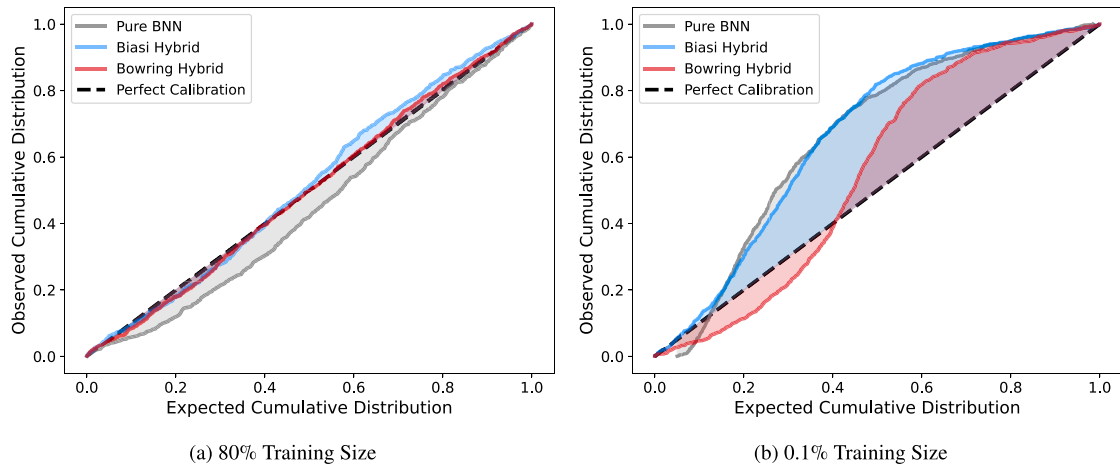


Fig. 7. Calibration curves for the BNN approach.

Table 8

Comparison of the DGP performance metrics between the pure and hybrid models.

Metric	Pure ML DGP	Biasi hybrid	Bowring hybrid
7350 training points (80% case)			
μ_{error}	6.348%	4.196%	5.171%
$\text{Max}_{\text{error}}$	84.09%	37.29%	62.17%
Mean rStd	2.008%	1.793%	3.605%
Max rStd	28.63%	18.59%	61.63%
rRMSE	9.413%	5.838%	7.680%
$F_{\text{error}} > 10\%$	16.76%	7.835%	11.97%
R^2	0.9780	0.9897	0.9833
9 training points (0.1% case)			
μ_{error}	62.92%	11.06%	10.32%
$\text{Max}_{\text{error}}$	881.5%	96.69%	88.76%
Mean rStd	9.839%	8.807%	7.362%
Max rStd	44.65%	681.9%	218.6%
rRMSE	100.9%	15.74%	15.28%
$F_{\text{error}} > 10\%$	87.49%	38.74%	35.26%
R^2	-0.0745	0.9537	0.9556

the smallest spread about its mean rStd value of 2.008%. Although the Biasi hybrid model shows a smaller mean relative standard deviation value compared with that of the pure ML model, its standard deviation for this distribution is larger. The Bowring hybrid model has the largest mean, standard deviation, and maximum. These models' trends are similar in the limited training data case but with larger means, maximums, and standard deviations.

The UQ calibration curves were then constructed. These curves are provided in Fig. 9. The Bowring hybrid model in both training data scenarios is the best calibrated compared with the other models, with miscalibration areas of 0.1063 in the plentiful case and 0.0944 in the limited case. All six of the cases between the two data scenarios have overconfident uncertainty estimates, leading to smaller-than-expected prediction intervals. By contrast, the pure ML models are the most poorly calibrated of the six DGP models (miscalibration areas of 0.1737 and 0.2085 for the plentiful and limited training scenarios, respectively). The Biasi hybrid model falls in between and also shows better calibration in the 0.1% scenario (miscalibration area of 0.1115) than in the 80% training scenario (miscalibration area of 0.1556).

4.4. Direct comparison between methods

With the results above, direct comparisons can be made across the ML techniques (DNN ensemble, BNN, DGP) rather than comparisons

simply being made between the pure and hybrid approaches performed within each. The first obvious metric to compare is the mean relative error. These values for the nine models of the 80% training set size case are visually compared in Fig. 10. Every model of the BNN and DGP has an increased error value compared with that of the ensemble; the DGP has the largest error of the three methods (6.348%). The DNN ensemble and BNN show relatively even performance between pure and hybrid models in this training size case, with the pure DGP model having error values a full percentage point above the closest hybrid model (6.348% compared with the Bowring hybrid's 5.171%). The DGP's pure ML model has the largest relative error of any model (over 4 percentage points above any DNN ensemble model), and the hybrids follow. All nine models, however, have a smaller error than when the bare Biasi and Bowring correlations are applied to the dataset. The DNN ensemble has the smallest error overall and consequently the largest difference compared with that of the bare correlations; its relative error values are below 2% (compared with the correlations' 6.935% and 6.778%).

The test set entries were then filtered using a "BWR" pressure range, set between 6900 kPa and 7200 kPa [1], to reduce the number of points for graphical analysis. This process was used for all the 80% and 0.1% training size cases. The remaining values were then plotted with prediction intervals created from their 2σ values to assess the uncertainty distribution while also considering parity. In a normal distribution, 2σ encloses roughly 95.4% of all values. Prediction intervals in this case are an estimate of the interval in which a new prediction of a given input combination will fall inside with a specified probability. Notably, these sigma values are of the same units of CHF and are not relative values as previously discussed. The first, Fig. 11, contains values for all models trained with the 80% size case. An initial visual inspection indicates a clear difference between the prediction intervals of the DNN ensemble and those of both the BNN and DGP methods in all cases. The DNN ensemble's bars are relatively small along the identity line, with gradually increasing magnitudes with increasing CHF and a few larger values in the central region. Additionally, there is a tight adherence of mean values to this line, as previously indicated by the high overall accuracy. Each of the three models of the DNN ensemble indicates comparable standard deviation values.

Both the BNN and DGP models have relatively higher amounts of variance in each of the points' prediction intervals, particularly of note in all cases of the BNN and in the Bowring model of the DGP hybrid cases. All three models of the BNN have prediction intervals of similar magnitude, with a general trend of increasing height as CHF increases. This nonuniform distribution in uncertainty is heteroscedastic and can be due to multiple factors, including data variability and potential systematic biases in the model's weights. Notably, the scales of both the x-axes and y-axes are different from those of the previous parity

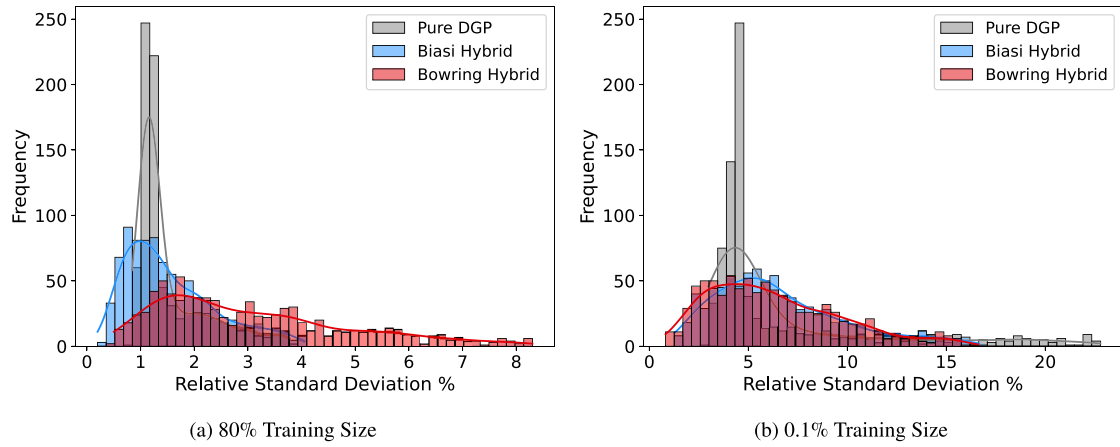


Fig. 8. Distributions of the rStd for the DGP approach: histograms and KDE curves. Outlier values are removed to improve readability.

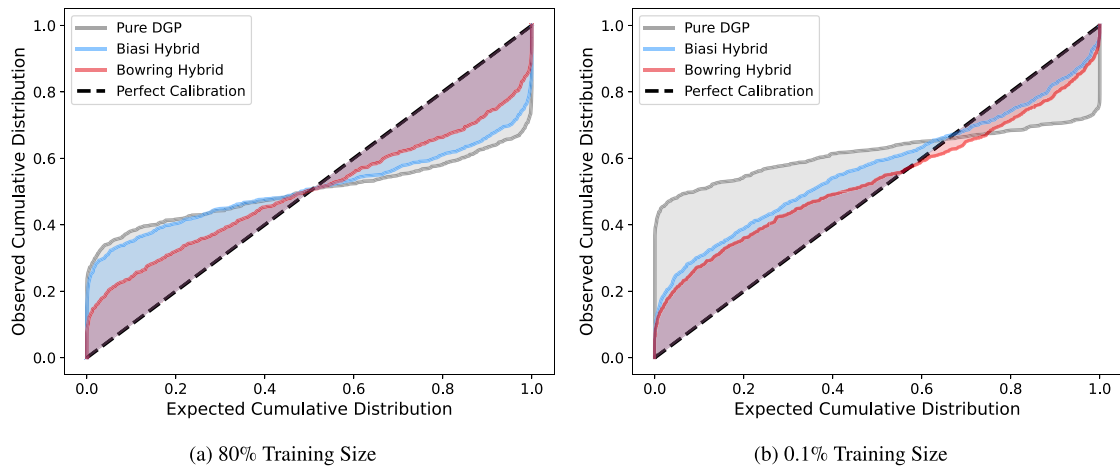


Fig. 9. Calibration curves for the DGP approach.

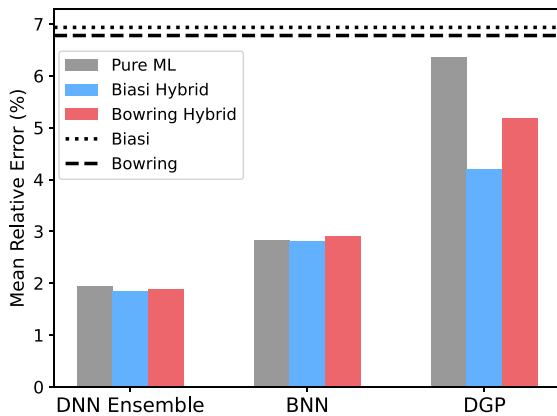


Fig. 10. Comparison of μ_{error} in the 80% training size case.

plots, making the 2σ values seem larger in comparison. Although the DGP models have larger error values compared with those of the BNN models, the DGP models' reported uncertainties are smaller in magnitude, with the largest located around the central region of the filtered CHF values.

The same process of constructing parity plots with prediction intervals was used for all models in the limited data scenario, which is

shown in Fig. 12. Each of the figures contains significantly different prediction interval behavior compared with those of Fig. 11. The purely data-driven ML models have intervals with widths several times those of the hybrid models. This is particularly of note in Fig. 12(b), which corresponds to the BNN cases. Every prediction made by the pure ML model is shown with 2σ widths on the order of 5000 kW m^{-2} , and all means are clustered about a single predicted value around 2000 kW m^{-2} . Both hybrid cases do not exhibit this behavior but rather follow the expected trend with means about the identity line and more reasonable uncertainties. The same effect is shown in Fig. 12(c) but with pure DGP uncertainties fluctuating greatly, even though the means predict about the same value. These behaviors, in addition to the increased error and uncertainties shown in the DNN ensemble's parity plot, support the conclusion that these pure ML approaches offer very little usefulness in cases of scarce training data. Notably, these uncertainty estimates are not meant to be compared between each method (DNN ensemble, BNN, DGP) because they are not well calibrated (as discussed in the previous subsections) and would not be deployed for real-world use.

Each of the three methods was assessed using three different analysis techniques: performance metrics/parity, uncertainty distribution, and calibrations. To implement one of these approaches in a production code, a prospective model must be highly accurate and interpretable and have small uncertainty estimates that are well calibrated. With these criteria, the DGP approaches can be quickly eliminated on the basis of their relatively high error (4.196% in the case of the Biasi

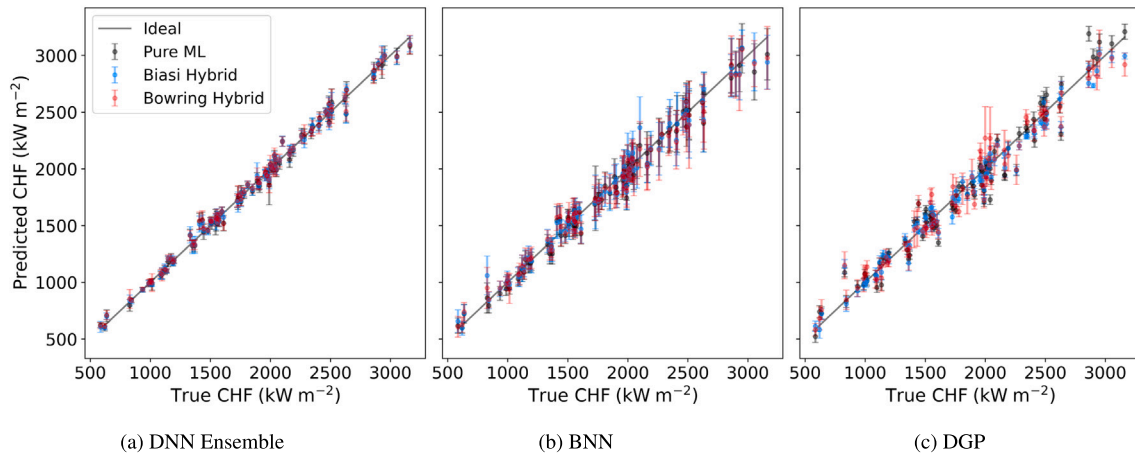


Fig. 11. Parity plots for BWR-range pressures comparing pure ML, Biasi hybrid, and Bowring hybrid across ML methods with 2σ error bars in the 80% training size case.

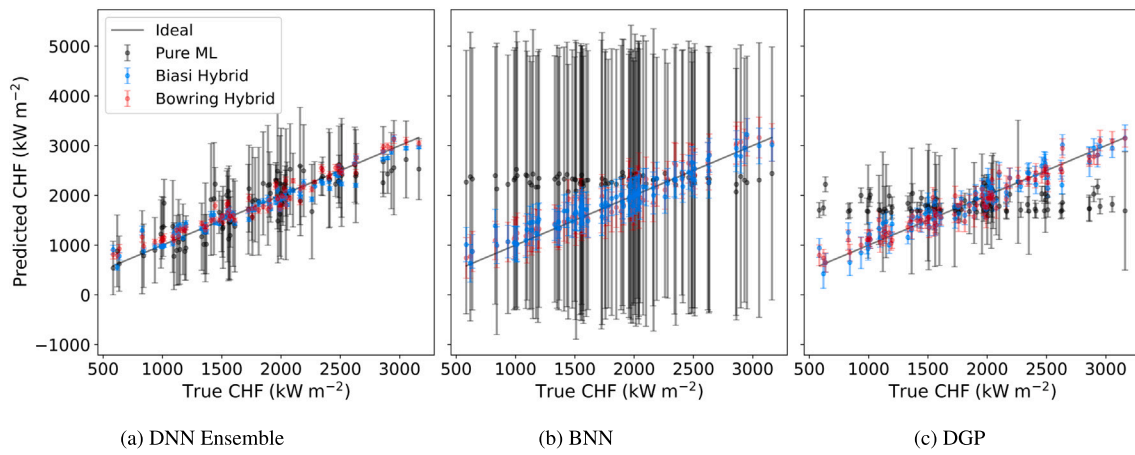


Fig. 12. Parity plots for BWR-range pressures comparing pure ML, Biasi hybrid, and Bowring hybrid across ML methods with 2σ error bars in the 0.1% training size case.

hybrid), in addition to their uncalibrated uncertainty behavior (despite small reported uncertainties). The BNN models, though well calibrated in UQ, have larger reported uncertainties, along with greater error metrics, in comparison with the DNN ensemble. The DNN ensemble in all three cases (pure ML and the correlation hybrids) obtained the most favorable performance metrics, in addition to small, well-calibrated uncertainties with a significantly tighter distribution compared with those of the BNN models. Although the DNN ensembles were shown to be slightly less calibrated with regard to the BNN models, they are still considered to be well calibrated. Of the DNN ensemble models, the hybrids are structurally the most interpretable, which is the final criterion for model selection. Because of these factors, the DNN-based Biasi hybrid model is a competitive candidate for future deployment in a production code.

5. Conclusions

This study developed and validated an uncertainty-aware hybrid modeling approach that combines ML with physics-based models to improve the prediction of CHF in nuclear systems. By training data-driven ML models to correct residuals from established empirical correlations using experimental data, the hybrid approach improves accuracy, reduces the potential for non-physical outputs, and increases interpretability through the constraints imposed by the base model.

All models were trained and tested on a subset of the public CHF dataset used to construct the 2006 Groeneveld LUT, restricted to dryout conditions as well as the correlations' ranges of validity. The Biasi and

Bowring correlations were used as base models in hybrid configurations with three different uncertainty-aware ML techniques: DNN ensembles, Bayesian neural networks, and deep Gaussian processes. A pure ML model configuration, without a base model, was also evaluated as a baseline. For each of these combinations, a plentiful training dataset size (7350 training points) and a limited scenario (9 training points) were used to assess the hybrid models' resistance to data scarcity.

The performance and uncertainty of the trained models were quantified with seven statistical measures and three analysis techniques: (1) parity plots to inspect accuracy, (2) relative standard deviation distributions to assess the behavior of uncertainty, and (3) calibration curves to ensure calibration of the reported uncertainties. These components enabled a comprehensive assessment of error and uncertainty across all model configurations.

The Biasi hybrid DNN ensemble exhibited the most favorable behavior in most measures, particularly in the plentiful training data scenario. This model outperformed the stand-alone (non-ML) Biasi correlation even in the limited data scenario, with only nine training points. The BNN-based models had slightly increased error and uncertainty metrics compared with those of the DNN ensemble, but with the best UQ calibration in the plentiful data case. The DGP-based models have the poorest performance in nearly all measures, with large UQ miscalibration errors. Both the pure BNN and DGP models broke down in the limited data scenario, collapsing to near-constant predictions for all input combinations. In all cases, the hybrid models outperformed the pure ML configurations. All models in the plentiful data scenario significantly outperformed the stand-alone Biasi

and Bowring correlations. However, in the limited data scenario, only the Bowring hybrid BNN model and the two hybrid DNN ensemble models obtained error metrics below the correlations.

Implementation of these UQ-enabled models within a real-world application, such as a subchannel code, would further demonstrate the feasibility of using these methods outside a pure research setting and provide a direct pathway for use in industrial applications. To be successfully deployed, the models must: (1) be lightweight to maintain the code's computational speed, (2) be general enough for use in diverse geometries/conditions, and (3) pass all relevant benchmarks as part of external validation. For preliminary investigation, the use of DNN variants is recommended for their comparatively low inference overhead, simplicity, and strong performance. While tubular channel models are important, they may not be immediately suitable for other geometries such as bundles where subchannels interact. The use of transfer learning techniques, such as parameter transfer and domain alignment, may offer a path to adapting the well-trained tube models to these alternate geometries.

Overall, the results of this study demonstrate that knowledge-based hybrid ML modeling approaches offer enhanced accuracy, interpretability, and greater resilience to data scarcity in the task of CHF prediction. Combined with uncertainty-aware methods, hybrid models are viable candidates for high-consequence applications in nuclear thermal hydraulics.

CRedit authorship contribution statement

Aidan Furlong: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Xingang Zhao:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Robert K. Salko:** Writing – review & editing, Supervision, Software, Project administration, Conceptualization. **Xu Wu:** Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported in part by an appointment to the US Department of Energy's Omni Technology Alliance Internship Program, sponsored by DOE and administered by the Oak Ridge Institute for Science and Education (ORISE). The authors from North Carolina State University were also funded by the DOE Office of Nuclear Energy Distinguished Early Career Program (DECP) under award number DE-NE0009467.

Data availability

Data will be made available on request.

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